

Wetlab immunogenicity assay to simulation-support bridge

HLA binding / stability assay

pMHC structure crosscheck; mutant pMHC 3x10ns; WT/decoy pMHC controls

pMHC multimer / tetramer with known TCR or T cells

TCR-pMHC 3x10ns; CDR3-peptide contacts; TCR unbinding PMF only for top cases

IFN-gamma ELISpot / ICS

TCR contact persistence, mutation-site contacts, WT/decoy specificity, kinetic proofreading proxy

Killing assay

Only upstream presentation/recognition plausibility plus WT/decoy risk